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Higgs Physics Potential of the FCC-μ^+e^- Collider at CERN

From Standard Model Benchmarks to Higgs Self-Couplings

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First scientific collaboration kick-off, 10 November 2026

Discussion participants: Krzysztof Piotrzkowski, Richard Ruiz, Piotr Kotko and Hamzeh Khanpour

Scientific Motivation and First-Meeting Objectives

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What we already have

- Full LO CC/NC H and HH matrices.
- 48 verified production configurations (1 and 2 TeV).
- Fiducial $HH \to 4b$, full $4b$, $b\bar{b}H$ and $ZH \to 4b$ studies.
- Original MadGraph Feynman-diagram exports.

What we need to decide

- Common SM input and collider baselines.
- κ_3–κ_{2V} interpretation.
- Detector/forward-acceptance priorities.
- Whether HHH is feasible at 1-2 TeV.
- First joint analysis and deliverable.

Physics takeaway

Use the completed SM calculations to define a credible coupling and experimental programme, not simply a catalogue of rates.

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Roadmap of the Scientific Discussion



1. The FCC-μe concept and FCC-ee complementarity.
2. Our process inventory, definitions and CERN workflow.
3. Inclusive H/HH: Richard's polarization study.
4. Original diagrams and Krzysztof's self-coupling question.
5. Earlier $4b$, associated-Higgs and muon-collider comparisons.
6. Higgs potential: κ_3, κ_{2V}, possible HHH.
7. Decisions, work packages and next calculations.

Physics takeaway

Numerical results first; coupling interpretation and collaboration decisions at the end.

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Why a High-Energy Positive-Muon–Electron Collider?

Distinctive initial state

- High-energy μ^+ collides with a FCC electron beam.
- $\sqrt{s} \sim 0.6$–2 TeV without same-flavour s-channel annihilation.
- WW/ZZ-mediated Higgs production at high Q^2.
- Tunable positive-muon helicity; lepton initial states without proton remnants.

Physics leverage

- Direct tree-level sensitivity to λ_3 through HH.
- Chiral electroweak structure test via polarization.
- Distinct CLFV and long-lived-particle reach.
- Companion intense neutrino beams.

Physics takeaway

Complementary to FCC-ee precision Higgsstrahlung; higher energy helps weak-boson fusion.

Beam Energies and Centre-of-Mass Reach

$$s \simeq 4E_{e^-}E_{\mu^+}, \qquad \sqrt{s} \simeq 2\sqrt{E_{e^-}E_{\mu^+}}$$

Two distinct benchmark families

Family	Case	E_{μ^+} [GeV]	E_{e^-} [GeV]	$\sqrt{s}$ [TeV]
FCC-μe machine-like	low	2000	120	0.9798
FCC-μe machine-like	high	5000	183	1.9131
Exact-energy scan	1 TeV	2000	125	1.0000
Exact-energy scan	2 TeV	5000	200	2.0000

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Luminosity, Staging and Running Assumptions

Revised proposal (arXiv v2), example 2 TeV μ^+ beam

FCC-ee electron energy	45.6 GeV	120 GeV
$\sqrt{s}_{\mu e}$	~ 0.60 TeV	~ 0.98 TeV
Estimated peak $\mathcal{L}$ [$10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$]	4.5	0.63
Collision rate	40 MHz	1 MHz

- Values are *proposal scenarios*, depending on source intensity, emittance and beam matching.
- v2 discusses multi-ab^{-1} possibilities integrated over FCC-ee stages; distribution among energies and polarization settings is not fixed.
- Stand-alone operation and the 5 TeV μ^+ extension remain design options.

Physics takeaway

Sensitivity estimates require $\mathcal{L}_{\mathrm{int}}$ at the exact energy and polarization; peak $\mathcal{L}$ is not integrated luminosity.

FCC-ee and FCC-μe: Complementary Higgs Programmes

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FCC-ee: precision factory

- e^+e^- at $\sqrt{s} \simeq 240$ and 365 GeV.
- ZH recoil method, Higgs couplings/width.
- Precision HZZ, HWW, heavy-flavour decays.
- Indirect self-coupling information through radiative corrections.

FCC-μe: high-energy complement

- WW/ZZ-fusion H and HH at $\sim$ 1–2 TeV.
- Direct tree-level HH dependence on λ_3.
- Polarization and $HHVV$ information.
- Potential combined Higgs-coupling / EFT interpretation.

Physics takeaway

Best physics case is *combined information*, not unsupported claims that either machine supersedes the other.

Detector Challenges from Highly Asymmetric Beams

Kinematic acceptance

- Large laboratory-frame boost along the energetic μ^+ beam.
- Forward charged leptons in NC production.
- Missing momentum for CC channels.
- $4b$ tagging in extended η acceptance.

Machine–detector interface

- Decay positrons from stored μ^+ beam.
- Nozzle shielding and beam separation.
- Trigger, calibration and backgrounds.
- Practical polarized-beam luminosity.

Physics takeaway

Generator cuts of $|\eta_b| < 5$ are not a detector design; reconstructible final states require dedicated simulation.

Complete Inventory of Investigated Higgs Processes

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FCC-μe completed

- CC: $\nu_e \bar{\nu}_\mu H$, $\nu_e \bar{\nu}_\mu HH$.
- NC: $e^- \mu^+ H$, $e^- \mu^+ HH$.
- Resonant $HH \to 4b$ and full $\nu\bar{\nu}4b$.
- NC $HH \to 4b$, $b\bar{b}H \to 4b$, full NC $4b$ and continuum diagnostic.

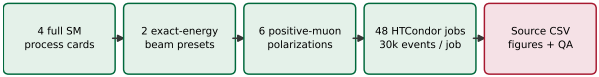
$\mu^+\mu^-$ comparison completed

- CC $HH \to 4b$; full $\nu\bar{\nu}4b$.
- NC associated $b\bar{b}H$.
- Resonant $ZH \to 4b$ at 1,2 TeV.

Not yet calculated

$\mu^+ e^- \to \nu_e \bar{\nu}_\mu HHH$ and
$\mu^+ e^- \to \mu^+ e^- HHH$,
anomalous-coupling scans.

CERN MadGraph Workflow and Reproducibility



Production

- MadGraph5_aMC@NLO 3.6.4, LO SM.
- Original process and parameter cards preserved.
- 48 exact-energy HTCondor jobs on CERN.
- 30,000 event requests per job; individual seeds.

Quality status

- 48/48 returned result records.
- Beam polarizations and energies audited.
- MC integration errors available.
- **Still needed:** independent seeds; forward NC stability.

Inclusive, Decayed and Fiducial Cross Sections

Dataset	Process definition	Selection / normalization
Inclusive H/HH	Stable H, full CC/NC LO matrix	No generation kinematic cuts; output σ_{prod}
Resonant $HH \to 4b$	Two $H \to b\bar{b}$ decays	$\lvert\eta_b\rvert < 5$; two 110–130 GeV pairs
NC $4b$	NC charged leptons	Above b cuts; $p_{T,\ell} > 1$ GeV at generation
Muon $ZH \to 4b$	Explicit $Z, H \to b\bar{b}$	$\lvert\eta_b\rvert < 5$; two 80–130 GeV pairs

Physics takeaway

Do not compare stable-H inclusive rates directly with fiducial four-bottom rates without the decay, cut and EW input conventions.

Polarization, Electroweak Inputs and Branching Fractions



Inclusive scan conventions

- MG5 beam 1: e^-, $P_{e^-} = 0$.
- MG5 beam 2: μ^+; $P_{\mu^+} = 0 \ldots 100\%$.
- Right-helicity μ^+ contributes to CC.
- Two separate exact-energy beam cards.

Historical $4b$ normalization

- Earlier resonant $HH \to 4b$ packages used $\mathrm{BR}(H \to b\bar{b}) = 0.844640$.
- Precision-SM nominal value is ~ 0.58.
- Ratios/absolute $4b$ yields are not interchangeable before reweighting or regeneration.

Physics takeaway

Check original parameter cards, resonance treatment, and branching fractions before quantitative sensitivity fits.

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Four Full-Matrix-Element Inclusive Higgs Channels

Channel	MadGraph5 generation command
CC H	`generate e- mu+ > ve vm~ h QCD=0`
CC HH	`generate e- mu+ > ve vm~ h h QCD=0`
NC H	`generate e- mu+ > e- mu+ h QCD=0`
NC HH	`generate e- mu+ > e- mu+ h h QCD=0`

Physics content

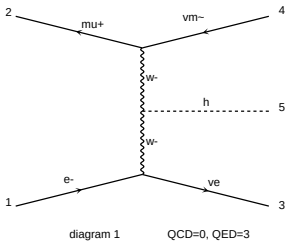
Full tree-level SM amplitudes without diagram exclusions; stable Higgs bosons; WW/ZZ fusion topology. HH includes $HHVV$, separate HVV, and $H^* \to HH$ contributions.

Physics takeaway

CC and NC probe complementary initial-state chiral currents and Higgs–gauge couplings.

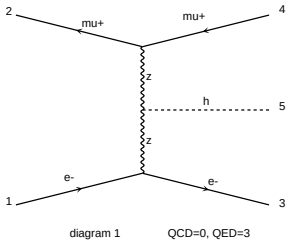
Single-Higgs Production: Original MadGraph Diagrams

CC: $\mu^+ e^- \to \bar{\nu}_\mu \nu_e H$



One exported diagram; *HWW* fusion coupling.

NC: $\mu^+ e^- \to \mu^+ e^- H$



One exported diagram; *HZZ* fusion coupling.

Physics takeaway

These are *original exported CERN MadGraph diagrams*, not generic schematic Feynman pictures.

CC Double Higgs: Four Original MadGraph Diagrams

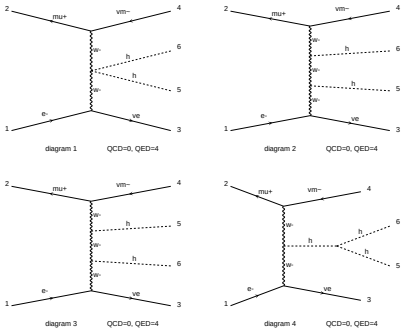


Diagram identification

- **1:** $HHWW$ contact.
- **2,3:** separate Higgs emissions through HWW couplings.
- **4:** $W^*W^* \to H^* \to HH$; trilinear HHH coupling.

Physics takeaway

The unrestricted CC HH process includes all four tree diagrams and their interference.

NC Double Higgs: Four Original MadGraph Diagrams

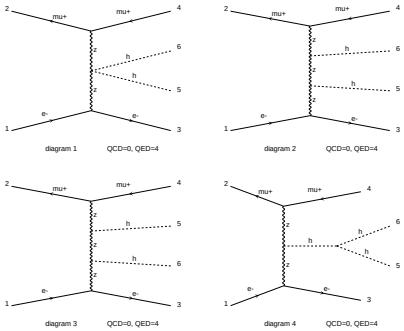


Diagram identification

- **1:** $HHZZ$ contact.
- **2,3:** separate Higgs emissions through HZZ couplings.
- **4:** $Z^*Z^* \to H^* \to HH$; trilinear HHH coupling.

Physics takeaway

The neutral-current amplitude includes contact, sequential and virtual-Higgs contributions coherently.

Krzysztof's Question: Are All HH Contributions Included?

Question addressed

Are both the Higgs self-coupling diagram $H^* \to HH$ and the separate HVV-emission graphs included?

$$\mathcal{M}_{HH} = \mathcal{M}_{HHVV} + \mathcal{M}_{\text{separate } HVV} + \mathcal{M}_{\lambda_3}$$

$$\sigma_{HH} \propto \left|\mathcal{M}_{HH}\right|^2$$

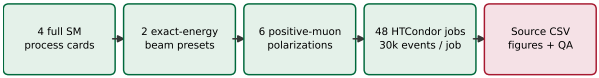
- **Yes:** original CC/NC MG5 exports contain diagrams 1–4, including virtual H^* exchange.
- Amplitude-level interference is retained; no diagram-specific cross sections were calculated.
- Arbitrary subsets of gauge-related diagrams need not be independently gauge invariant.

Physics takeaway

The existing SM baselines contain λ_3; a coupling scan is needed to quantify sensitivity.

Richard's Six-Point Polarization Task: 48 CERN Runs

$4 \times 2 \times 6 = 48$ **production configurations**



Input scan

- 4 channels: CC/NC H, HH.
- Exact $\sqrt{s} = 1, 2$ TeV.
- $P_{\mu^+} = 0, 20, 40, 60, 80, 100\%$.

CERN output

- 30,000 events per configuration.
- 48/48 archived cross-section records.
- Run-card and polarity QA passed; physics convergence ongoing.

Inclusive Higgs Cross Sections at 1 TeV

Exact $\sqrt{s} = 1$ TeV; $e^-(125)$ GeV $+ \mu^+(2000)$ GeV; $P_{e^-} = 0$

Each entry: $\sigma \pm \delta\sigma_{\mathrm{MC}}$ [fb].

P_{μ^+}	CC H	CC HH	NC H	NC HH
0%	210.540 ± 0.209	0.074384 ± 0.00013	20.835 ± 0.025	0.0096201 ± 0.0000146
20%	252.710 ± 0.158	0.088979 ± 0.00015	21.770 ± 0.015	0.010045 ± 0.0000167
40%	294.730 ± 0.271	0.104016 ± 0.00019	22.660 ± 0.014	0.010487 ± 0.0000160
60%	336.830 ± 0.199	0.118825 ± 0.00021	23.611 ± 0.019	0.010895 ± 0.0000148
80%	378.570 ± 0.243	0.133230 ± 0.00023	24.476 ± 0.015	0.011308 ± 0.0000174
100%	421.150 ± 0.433	0.148239 ± 0.00024	25.402 ± 0.018	0.011723 ± 0.0000180

Physics takeaway

At $P_{\mu^+} = 80\%$: CC $HH = 0.13323$ fb and NC $HH = 0.011308$ fb; stable Higgs, not $4b$.

Inclusive Higgs Cross Sections at 2 TeV

Exact $\sqrt{s} = 2$ TeV; $e^-(200)$ GeV $+ \mu^+(5000)$ GeV; $P_{e^-} = 0$

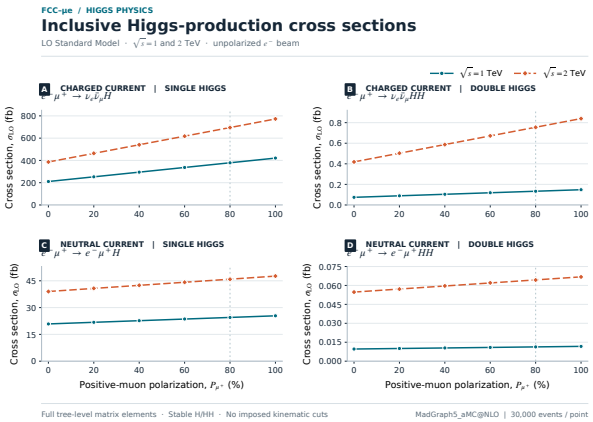
Each entry: $\sigma \pm \delta\sigma_{\mathrm{MC}}$ [fb].

P_{μ^+}	CC H	CC HH	NC H	NC HH
0%	386.010 ± 0.303	0.418510 ± 0.00066	38.991 ± 0.030	0.054712 ± 0.0000851
20%	463.490 ± 0.399	0.502952 ± 0.00085	40.760 ± 0.027	0.057089 ± 0.0000871
40%	540.620 ± 0.371	0.587089 ± 0.00100	42.460 ± 0.045	0.059636 ± 0.0000859
60%	617.270 ± 0.428	0.671917 ± 0.00115	44.171 ± 0.042	0.062066 ± 0.0000868
80%	695.000 ± 0.614	0.754495 ± 0.00128	45.829 ± 0.055	0.064299 ± 0.0000957
100%	772.690 ± 0.517	0.839521 ± 0.00140	47.642 ± 0.033	0.066770 ± 0.00010

Physics takeaway

CC HH: 0.41851 (P0) $\rightarrow$ 0.75450 (P80) $\rightarrow$ 0.83952 fb (P100).

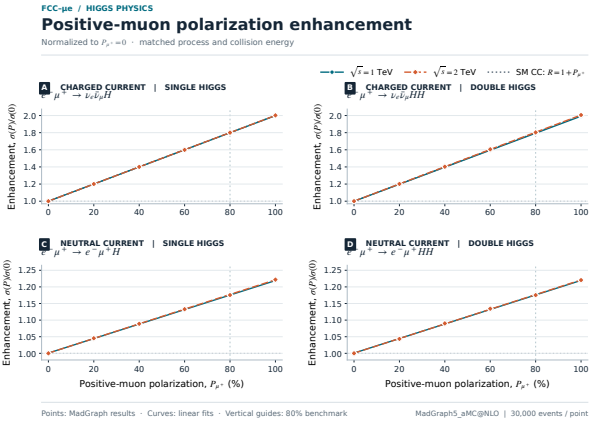
Absolute Higgs Cross Sections versus Muon Polarization



Physics takeaway

CC rates are larger; the higher energy strongly benefits the *HH* channel. Error bars are MC integration errors.

Polarization Enhancement Ratios: CC versus NC



Physics takeaway

$R_{CC}(80\%) \simeq 1.8$, while $R_{NC}(80\%) \simeq 1.175$ for both H and HH.

Polarization Slopes and Linearity Tests

$$\sigma(p) = a(1 + Ap), \qquad p = P_{\mu^+}/100$$

	$A(1\ \mathrm{TeV})$	$A(2\ \mathrm{TeV})$	$\chi^2/4$ (1 TeV)	$\chi^2/4$ (2 TeV)
CC H	0.9982 ± 0.0020	1.0007 ± 0.0018	$1.77/4$	$2.38/4$
CC HH	0.9934 ± 0.0040	1.0053 ± 0.0039	$2.84/4$	$0.94/4$
NC H	0.2181 ± 0.0012	0.2208 ± 0.0011	$5.08/4$	$3.40/4$
NC HH	0.2186 ± 0.0022	0.2204 ± 0.0022	$1.93/4$	$3.36/4$

Interpretation

$A_{CC} \simeq 1$ (chiral charged weak current); $A_{NC} \simeq 0.22$ (parity-violating Z couplings). Fits test generator behavior, not Higgs-coupling measurement precision.

Chiral Electroweak Interpretation of Polarization

Charged current

$$R_{CC}(p) \simeq 1 + p$$

Only the appropriate high-energy μ^+ helicity enters the W current.
$R_{CC}(+80\%) \simeq 1.8$.

Neutral current (pure-Z limit)

$$g_L = -\tfrac{1}{2} + s_W^2, \quad g_R = s_W^2$$

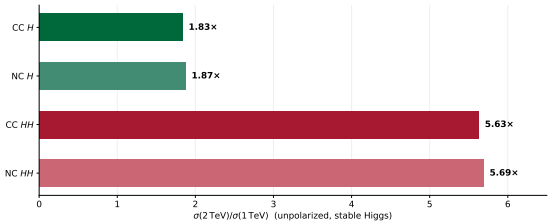
$$A_{NC} \simeq \frac{g_L^2 - g_R^2}{g_L^2 + g_R^2}$$

Both chirality states contribute;
$A_{NC} \approx 0.22$.

Physics takeaway

The NC analytic comparison depends on the exact electroweak input scheme; polarization alone is not a weak-angle precision measurement.

Energy Scaling and Illustrative Higgs Event Rates



Hypothetical $\mathcal{L} = 3$ ab^{-1}

At $P_{\mu^+} = 80\%$:

CC *HH*, 1 TeV: ~ 400 produced events.

CC *HH*, 2 TeV: ~ 2263 produced events.

NC *HH*, 2 TeV: ~ 193 produced events.

Physics takeaway

These are raw inclusive yields: *before* Higgs decays, tagging, acceptance, backgrounds or realistic luminosity allocation.

CC Double Higgs to $4b$ and Complete $\nu\bar{\nu}4b$

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FCC-μe

$$\mu^+ e^- \to \nu_e \bar{\nu}_\mu HH \to \nu_e \bar{\nu}_\mu 4b$$

$$\mu^+ e^- \to \nu_e \bar{\nu}_\mu b\bar{b}b\bar{b}$$

Machine-like $\sqrt{s} = 0.980, 1.913$ TeV.

Muon collider

$$\mu^+ \mu^- \to \nu_\ell \bar{\nu}_\ell HH \to \nu_\ell \bar{\nu}_\ell 4b$$

$$\mu^+ \mu^- \to \nu_\ell \bar{\nu}_\ell 4b$$

$\ell = e, \mu, \tau$ (summed). $\sqrt{s} = 1, 2$ TeV.

Physics takeaway

Resonant HH samples are distinct from the full $4b$ matrix element, which already contains Higgs and continuum contributions.

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Four-Bottom Reconstruction and Fiducial Selection

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Charged-current $HH \to 4b$ and full $\nu\bar{\nu}4b$

$$|\eta_{b_i}| < 5 \quad (i = 1, \ldots, 4), \qquad p_{T,b}^{\min} = 0$$

$$110 < m_{b\bar{b}}^{(1)} < 130\,\mathrm{GeV}, \qquad 110 < m_{b\bar{b}}^{(2)} < 130\,\mathrm{GeV}$$

- Two disjoint $b\bar{b}$ pairs; source implementation uses allowed charge-opposite pairing assignments.
- Observables defined using *partons*, not clustered jets: no shower, tagging or detector effects.
- NC samples additionally regulate the outgoing charged leptons with $p_{T,\ell} > 1$ GeV at generation.
- The later $\mu^+\mu^- \to ZH \to 4b$ study uses **80–130 GeV** windows instead.

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Fiducial CC HH and $4b$: FCC-μe vs Muon Collider

LO parton-level fiducial cross sections [fb]

Collider	$\sqrt{s}$	$HH \to 4b$: P0 / P80	Complete $\nu\bar{\nu}4b$: P0 / P80
FCC-μe	0.980 TeV	0.04912 / 0.08816	0.05355 / 0.09195
$\mu^+\mu^-$	1.000 TeV	0.07047 / 0.11582	0.07530 / 0.12472
FCC-μe	1.913 TeV	0.26478 / 0.47611	0.28157 / 0.50745
$\mu^+\mu^-$	2.000 TeV	0.30624 / 0.54606	0.32480 / 0.57873

Interpretation requires consistent normalisation

Earlier resonant $HH \to 4b$ used $\mathrm{BR}(H \to b\bar{b}) = 0.844640$; these historical fiducial rates should not be treated as conventional precision-SM $4b$ yields.

Resonant HH, Nonresonant $4b$ and Interference

$$\mathcal{M}_{\text{full 4b}} = \mathcal{M}_{HH} + \mathcal{M}_{\text{single H}} + \mathcal{M}_{\text{nonres}} + \cdots$$

Different generator objects

- Resonant HH decay chains impose intermediate resonances.
- Full $\nu\bar{\nu}4b$ includes allowed trees and interference.
- Single-resonant and continuum pieces can overlap.

Danger of naive subtraction

$\sigma_{\text{full}} - \sigma_{HH}$ is *not* generically an independently simulated, gauge-invariant physical background: interference, off-shellness, generator settings and resonance definitions matter.

Physics takeaway

Next step: common-generation comparison with resonance-aware integration and detector-driven regions.

Neutral-Current Four-Bottom Channel Definitions

Distinct NC generator samples at $\sqrt{s} \simeq 0.980$ TeV

1. $e^-\mu^+ \to e^-\mu^+ HH \to e^-\mu^+ 4b$ (double Higgs).
2. $e^-\mu^+ \to e^-\mu^+ b\bar{b}H \to e^-\mu^+ 4b$ (associated Higgs).
3. $e^-\mu^+ \to e^-\mu^+ b\bar{b}b\bar{b}$ (complete final-state amplitude).
4. Same $4b$ final state with internal Higgs graphs excluded (continuum diagnostic).

- Outgoing-lepton generation cut: $p_{T,\ell} > 1$ GeV.
- Four b partons with $|\eta_b| < 5$ and two 110–130 GeV pairs.
- Samples are *not* mutually exclusive event classes; graph exclusions are not automatically gauge invariant.

NC Four-Bottom Cross Sections and Their Limitations

Process at $\sqrt{s} = 0.980$ TeV	σ_{fid} [fb]	Comment
NC $HH \to 4b$	0.0063824	Resonant HH
NC $b\bar{b}H \to 4b$	0.0129250	Overlaps HH-sensitive phase space
Complete NC $4b$	0.0077129	442/30,000 events pass
H-excluded continuum	0.0008253	88/50,000 pass

Outstanding numerical consistency issue

Full $4b$ fiducial rate (0.00771 fb) lies below the separately generated $b\bar{b}H$ rate (0.01293 fb): this is **not explained** by the archived audit. Do not attribute it to interference without validation.

Physics takeaway

Priority: resonance-aware integration, independent seeds and independent $4b$ pairing/replay.

Associated $b\bar{b}H$ at Higher Energy and Polarization

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Selected preliminary $+80\%$ μ^+ results

Collider	$\sqrt{s}$ [TeV]	σ_{fid} [fb]
FCC-μe	1.913	0.081843
$\mu^+\mu^-$	2.000	0.094913

- $b\bar{b}H \to 4b$ with outgoing charged leptons; distinct from $\nu\bar{\nu}HH$.
- Exact generator definitions, selection rules and available statistics must be aligned before precision comparison.
- May contain singly and doubly Higgs-resonant regions, depending on the process definition.

Physics takeaway

Associated-Higgs benchmarks are useful for background strategy, but these rates are *preliminary*.

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Muon-Collider $ZH \to 4b$: Distinct Four-Bottom Topology

Explicit MadGraph resonant decay chain

```
generate mu+ mu- > z h QCD=0 QED=2, z > b b~, h > b b~
```

$\sqrt{s}$	σ_{before} [fb]	σ_{fid} [fb]
1 TeV	1.564500	1.527161 ± 0.001589
2 TeV	0.378880	0.369294 ± 0.000408

- Four b partons, $|\eta_b| < 5$, two pairs in **80–130 GeV**; $p_T^{\min}(b) = 0$.
- Includes $Z \to b\bar{b}$ and $H \to b\bar{b}$ decays; no extra BR factors.
- Ordinary s-channel $\mu^+\mu^- \to Z^* \to ZH$ has no μ^+e^- counterpart with flavour-conserving SM couplings.

What the Four-Bottom Programme Establishes

Reliable building blocks

- Fiducial $HH \to 4b$ cross sections.
- Complete $\nu\bar{\nu}4b$ amplitude comparisons.
- NC associated-Higgs and full $4b$ samples.
- Resonant muon-collider $ZH \to 4b$ reference.

Unresolved before sensitivities

- Earlier Higgs BR normalization.
- NC full versus $b\bar{b}H$ rate inconsistency.
- Physical acceptance of very forward particles.
- Detector response and multijet backgrounds.
- Independent seeds and NLO corrections.

Physics takeaway

These are production-level benchmarks and a background-analysis starting point, not a validated λ_3 sensitivity forecast.

Higgs Self-Couplings and the κ_3–κ_{2V} Programme

$$V(h) = \frac{m_h^2}{2}h^2 + \frac{m_h^2}{2v}h^3 + \frac{m_h^2}{8v^2}h^4$$

$$\kappa_3 = \frac{\lambda_3}{\lambda_3^{\mathrm{SM}}}, \qquad \kappa_{2V} = \frac{g_{HHVV}}{g_{HHVV}^{\mathrm{SM}}}$$

Why HH constrains the potential

- $VV \to H^* \to HH$ contains λ_3.
- $HHVV$ contact and separate HVV terms interfere.
- At fixed other couplings: $\sigma(\kappa_3) = c_0 + c_1\kappa_3 + c_2\kappa_3^2$.

Proposed scan

- Common SM input, consistent HEFT/SMEFT model.
- κ_3–κ_{2V} grid at 1/2 TeV.
- Charged and neutral currents, with polarization.

Physics takeaway

No κ_3 or κ_{2V} limit has yet been extracted from our 48 SM-rate calculations.

Differential Observables, EFT and Parameter Degeneracies

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Observables to generate

- m_{HH}, especially threshold versus high-mass tail.
- $p_{T,H}$ and Higgs rapidities.
- Higgs opening angle and $\cos\theta$.
- Outgoing lepton p_T, η in NC.
- $\not{p}_T$ and $4b$ reconstruction for CC.

Interpretation strategy

- HEFT: flexible Higgs-coupling modifiers.
- SMEFT: gauge-invariant operator correlations, validity domain.
- Joint CC+NC+FCC-ee inputs for coupling degeneracies.
- Shape likelihood with realistic backgrounds.

Physics takeaway

The HH threshold and high-m_{HH} regions can respond differently to κ_3 and κ_{2V}.

Beyond Double Higgs: Triple-Higgs Production

Two proposed full SM MadGraph reactions (not yet executed)

```
generate e- mu+ > ve vm~ h h h QCD=0
generate e- mu+ > e- mu+ h h h QCD=0
```

- Direct LO contributions involving λ_4, λ_3 and Higgs–gauge couplings.
- Test $\sqrt{s} = 1, 2$ TeV at $P_{\mu^+} = 0, 80, 100\%$.
- Validate integration near $m_{HHH} \geq 375$ GeV threshold.
- Evaluate $\sigma_{HHH} \times \mathcal{L}$ before decay/cuts.
- Explore $6b$, $4b2\tau$ and $4b2\gamma$ modes.
- If rates are tiny, test higher-energy exploratory points separately.

Physics takeaway

No *HHH* cross-section numbers or κ_4 sensitivities have been calculated; feasibility comes first.

Proposed Collaboration Work Packages

Work package	Immediate deliverable	Discussion
WP1: baseline QA	common parameter cards, cut definitions, seed reruns	inputs / owners
WP2: Higgs EFT	tested κ_3–κ_{2V} UFO/SMEFT framework	basis / validity
WP3: shapes	m_{HH}, p_T, angles, CC/NC polarization grids	observables
WP4: experiment	$HH \to 4b$ decays, b tagging, NC lepton acceptance, backgrounds	detector scenario
WP5: triple Higgs	first SM HHH cross sections and event-yield decision	feasibility
WP6: context	FCC-ee synergy, novelty vs earlier $e\mu$ HH literature	paper strategy

Physics takeaway

Workstream leadership and deadlines to be agreed by the four participants during this meeting.

Summary: Results Established and Decisions for the Kick-off

Established preliminary results

- 48 CC/NC H/HH inclusive LO points.
- CC $P_{\mu^+} = 80\%$: $\sim 1.8\times$ enhancement; NC $\sim 1.175\times$.
- HH production grows $\sim 5.6\times$ from 1 to 2 TeV.
- Original MG5 $HHVV$, separate-emission and $H^* \to HH$ diagrams verified.
- Fiducial $4b$ / ZH comparisons available with caveats.

Decisions requested

1. Common machine/physics benchmarks?
2. Consistent Higgs branching fractions?
3. HEFT or SMEFT choice?
4. Priority κ_3–κ_{2V} observables?
5. First detector/4b and HHH tests?
6. Joint-paper scope and milestone?

Physics takeaway

Can FCC-μe deliver a *distinctive combined* Higgs-coupling programme alongside FCC-ee?


THANK
YOU!

B1: FCC-μe Machine Working Points (Proposal v2)

Proposal examples: 2 TeV μ^+ beam (revised arXiv v2)

Quantity	e^- at 45.6 GeV	e^- at 120 GeV
$\sqrt{s}_{\mu e}$	$\sim$ 0.60 TeV	$\sim$ 0.98 TeV
Peak $\mathcal{L}$ [$10^{34}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$]	4.5	0.63
Collision rate [MHz]	40	1
β^*_{x,e^-} [m]	0.24	0.08
β^*_{y,e^-} [mm]	3.5	2.3
μ^+ bunch population [10^{10}]	0.2	1.0

- 5 TeV μ^+ stage and standalone running are exploratory proposal options.
- Design feasibility depends on source yield, emittance preservation, levelling and MDI.

B2: Process Generator Syntax and Analysis Inventory

Exact inclusive MG5 commands

```
generate e- mu+ > ve vm~ h QCD=0
generate e- mu+ > ve vm~ h h QCD=0
generate e- mu+ > e- mu+ h QCD=0
generate e- mu+ > e- mu+ h h QCD=0
```

Other source-defined final states

FCC-μe: $\nu_e\bar{\nu}_\mu HH \to \nu_e\bar{\nu}_\mu 4b$, full $\nu_e\bar{\nu}_\mu 4b$; NC $e^-\mu^+HH \to e^-\mu^+ 4b$, $e^-\mu^+ b\bar{b}H \to 4b$, full NC $4b$ and Higgs-excluded diagnostic.
$\mu^+\mu^-$: $\nu_\ell\bar{\nu}_\ell HH \to \nu_\ell\bar{\nu}_\ell 4b$, full $\nu_\ell\bar{\nu}_\ell 4b$, $b\bar{b}H$, and $ZH \to 4b$.

Physics takeaway

Exact historical generator restrictions and cuts are defined in original archives; do not infer unsupplied syntax.

B3: Complete 1 TeV Inclusive Table with MC Errors

$\sigma \pm \delta\sigma_{\mathrm{MC}}$ in fb, exact $\sqrt{s} = 1$ TeV.

P_{μ^+}	CC H	CC HH	NC H	NC HH
0%	210.540 ± 0.209	0.074384 ± 0.00013	20.835 ± 0.025	0.0096201 ± 0.0000146
20%	252.710 ± 0.158	0.088979 ± 0.00015	21.770 ± 0.015	0.010045 ± 0.0000167
40%	294.730 ± 0.271	0.104016 ± 0.00019	22.660 ± 0.014	0.010487 ± 0.0000160
60%	336.830 ± 0.199	0.118825 ± 0.00021	23.611 ± 0.019	0.010895 ± 0.0000148
80%	378.570 ± 0.243	0.133230 ± 0.00023	24.476 ± 0.015	0.011308 ± 0.0000174
100%	421.150 ± 0.433	0.148239 ± 0.00024	25.402 ± 0.018	0.011723 ± 0.0000180

Source-validation fields

24 unique entries; all status `COMPLETED_UNVALIDATED`; each requests 30,000 events. Independent seeds and relative MC errors are in `data/cross_sections_48.csv`.

Seed convention: source seeds are linked to the unique process/energy/polarization record; no common-seed substitution.

B4: Complete 2 TeV Inclusive Table with MC Errors

$\sigma \pm \delta\sigma_{\mathrm{MC}}$ in fb, exact $\sqrt{s} = 2$ TeV.

P_{μ^+}	CC H	CC HH	NC H	NC HH
0%	386.010 ± 0.303	0.418510 ± 0.00066	38.991 ± 0.030	0.054712 ± 0.0000851
20%	463.490 ± 0.399	0.502952 ± 0.00085	40.760 ± 0.027	0.057089 ± 0.0000871
40%	540.620 ± 0.371	0.587089 ± 0.00100	42.460 ± 0.045	0.059636 ± 0.0000859
60%	617.270 ± 0.428	0.671917 ± 0.00115	44.171 ± 0.042	0.062066 ± 0.0000868
80%	695.000 ± 0.614	0.754495 ± 0.00128	45.829 ± 0.055	0.064299 ± 0.0000957
100%	772.690 ± 0.517	0.839521 ± 0.00140	47.642 ± 0.033	0.066770 ± 0.00010

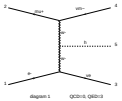
Source-validation fields

24 unique entries; all status `COMPLETED_UNVALIDATED`; each requests 30,000 events. Original run IDs/seeds and errors are preserved in packaged CSV and CERN JSON.

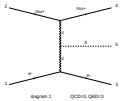
Clarification: this exact 2 TeV benchmark differs from historical FCC-μe $\sqrt{s} = 1.913$ TeV.

B5: Original MG5 Diagrams and Source Identification

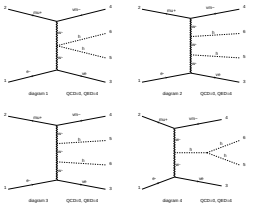
CC H (1)



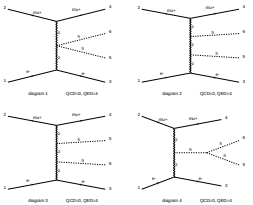
NC H (1)



CC HH (4)



NC HH (4)



Physics takeaway

Original diagram numbers from CERN exports retained. HH diagrams 1 contact; 2/3 separate emission; 4 virtual $H^* \to HH$.

B6: Fiducial Selection and Higgs-BR Normalization

Historical sample	Source cuts	Critical normalization / overlap
CC $HH \to 4b$ / full $4b$	$\lvert\eta_b\rvert < 5$; two 110–130 GeV pairs; $p_T^{\min}(b) = 0$	$\mathrm{BR}(H \to b\bar{b}) = 0.844640$ for decays
NC $HH/b\bar{b}H$/full $4b$	CC b cuts + outgoing $p_{T,\ell} > 1$ GeV generation	Overlapping amplitudes; full NC $4b$ still under review
Muon-collider $ZH \to 4b$	$\lvert\eta_b\rvert < 5$; two 80–130 GeV pairs; $p_T^{\min}(b) = 0$	Z and H branching already included in generator rate

Physics takeaway

Do not equate generator-level partons with tagged jets; normalize common decay modes before projections.

B7: Reproducibility, Integration and Open QA Checks

Verified in archives

- 48 unique rows: 4 channels $\times$ 2 $\sqrt{s}$ $\times$ 6 P.
- All returned jobs recorded with success artifacts.
- MG5 beam 1/2 energies and P auditable.
- Original process cards and selected run cards preserved.
- Original diagrams independently inspected.

Not independently validated

- Multiple random-seed consistency.
- NC forward-lepton cut/regulator stability.
- Full NC $4b$ versus $b\bar{b}H$ inconsistency.
- Decay normalization across old packages.
- Detector/beam-background and higher-order theory.

Physics takeaway

MC integration uncertainty is not the total theory or detector uncertainty.

B8: Primary References and Scientific Context

- K. Piotrzkowski, *FCC-μe: High Energy, High Luminosity Muon–Electron Collider & Neutrino Factory at CERN*, arXiv:2610.40451v2 (2026).
- FCC Collaboration, *Future Circular Collider Feasibility Study Report, Vol. 1*, Eur. Phys. J. C **85**, 1468 (2025), doi:10.1140/epjc/s10052-025-15077-x.
- Q.-H. Cao, K. Cheng, Y. Liu and X.-R. Wang, *Measuring hhWW coupling at lepton colliders*, Phys. Rev. D **109**, 073005 (2024).
- H. Khanpour, *CERN internal LO process, run-card, numerical and diagram audits*, October–November 2026 (provided collaboration archives).
- J. Alwall et al., *The automated computation of tree-level and next-to-leading-order differential cross sections*, JHEP **07**, 079 (2014), arXiv:1405.0301.

Physics takeaway

Collider projection, literature sensitivity and original CERN numerical calculations are differentiated throughout this presentation.